

Summary

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- **The Chain Rule:** If $f(x)$ and $g(x)$ are differentiable functions, then
$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$
- **The Derivatives of Trigonometric Functions:**

$$\begin{array}{ll}\frac{d}{dx} \sin(x) = \cos(x) & \frac{d}{dx} \cos(x) = -\sin(x) \\ \frac{d}{dx} \tan(x) = \sec^2(x) & \frac{d}{dx} \cot(x) = -\csc^2(x) \\ \frac{d}{dx} \sec(x) = \sec(x) \tan(x) & \frac{d}{dx} \csc(x) = -\csc(x) \cot(x)\end{array}$$